AKDENIZ UNIVERSITY · FACULTY OF ENGINEERING · DEPARTMENT OF COMPUTER ENGINEERING

CSE 491 — Project Proposal

Project title	A Human-in-the-Loop Knowledge Graph Generation Engine for Preventing Hallucination and Schema Drift in LLMs: An Application on Hadith Isnad Chains
Group no.	
Project supervisor	Res. Asst. Dr. Taha Yiğit Alkan
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Group members

#	Name and surname	Student no.	Role / responsibility in the project
1	Yahya Efe Kuruçay		Architecture, validation checks, decision record, scoring script
2	Deniz Büyükşahin		Labelling guideline, annotation (annotator 1), review screen
3	Alperen Ulukaya		Isnad segmenter, annotation (annotator 2), chain view

1. Summary

Large language models (LLMs) can turn text into knowledge graph facts, but they add facts the source does not contain and drift away from the schema they were given. GiraffeGraph is an engine in which the model only proposes. Each proposed fact carries a quotation from the source, is checked by deterministic rules, and reaches the graph only through a recorded human decision or a recorded acceptance policy. The design rests on one claim: what such a system has to store is the decision of a person, and what can be derived from the source is derived again.

Our case study is the hadith isnad, the chain of narrators that precedes a hadith text. In a first measurement on 300 hadiths, a check of whether the extracted name occurs in the model's own quotation flagged 122 of the 145 proposals that disagreed with existing platform labels. Those labels were not checked by people. By the end of CSE 491 we will deliver a working prototype that processes Arabic hadiths end to end and reports how many errors each check catches on 50 hadiths labelled by two group members, with disagreements decided by three hadith scholars. Entity resolution of narrators, the larger experiment and the question-answering layer are planned for Senior Design Project II.

2. Problem Definition and Motivation

When an LLM extracts a narrator chain from a hadith, four things go wrong. It outputs a narrator the text does not contain, or silently rewrites a name it did find. The same relation appears under different names in different runs (schema drift). One narrator is written in several ways and several narrators share one short name. After the graph is built, nobody can tell which sentence a fact came from or who accepted it.

This matters because the reliability of a hadith is judged from its chain. One wrong narrator changes the verdict, so a graph that is mostly right is not usable by a scholar.

Target users. Researchers and students in hadith studies, and engineers who build knowledge graphs whose facts must be traceable to their source.

What we already observed. On 12 October 2026 we ran one model (Claude Haiku 5.5) on 300 hadiths, 50 from each of the six canonical collections, and scored the output against the narrator labels of the Ifta Sunnah platform (1,778 reference mentions, loose matching):

- F1 was 0.923 for a rule-based parser, 0.890 for the LLM given only the schema and 0.956 for the LLM given an isnad-aware prompt, so raw extraction accuracy is not where we can contribute most.
- Checking that the model's quotation occurs in the source flagged nothing. Checking that the extracted name occurs inside the model's own quotation flagged 155 of 1,688 proposals (9.2%), including 122 of the 145 that disagreed with the labels, with 33 false alarms. Most flagged proposals were correct names that the model had transliterated or changed in grammatical case, not invented narrators. Precision of the proposals accepted without review rose from 0.914 to 0.985.

- The model's self-reported confidence separated right from wrong proposals poorly (AUC 0.634). Sending every proposal below 0.85 to review would send 16.3% of them and catch 50 of the 145.
- In the same hadiths, 10.0% of the 2,150 chain positions use a name shared by several of the 740 persons, and 25.8% of the persons appear under more than one name.

The labels are not a human gold standard, so these are comparisons between methods. One aim of the project is to repeat them on hand-labelled data.

The claim behind the design. A candidate that a parser or a model extracts can be extracted again, so it is checked, not trusted, and it does not have to be kept. An acceptance, a correction or the identification of a narrator by a person cannot be derived from any text and has to be stored once, with who decided and when. Two exploratory simulations of 13 October 2026, with no person taking part, looked at this. On 27,056 labelled chains of AR-Sanad [5], an identity decision reused whenever the same written name appears was wrong for one mention in ten; narrowing the decision only after a rule on generations and death years found a contradiction lowered the silent error to 0.014 per mention. In a second domain, a coding agent answered why-questions about the engine's code 0% to 4% correctly from the code alone and 88% to 100% correctly with a ledger of decisions (12 questions, two runs each).

3. Objectives, Scope and Success Criteria

Objectives

- Build an engine that turns Arabic hadith text into a narrator-chain graph in which every fact is tied to a source quotation and to a recorded decision. It covers loading hadiths from Sanadset 650K [1], rule-based separation of isnad and text, one LLM call per chain constrained to a fixed schema, three deterministic checks (the type is in the schema; the quotation occurs in the source; the value occurs in the quotation), one domain rule (the Prophet is not a narrator), a review screen and a canonical graph.
- Measure how many extraction errors each check catches on its own against hand-labelled data, and compare it with the model's own confidence when the same share of proposals is sent to review.
- Build a reference set of 50 hadiths labelled by two annotators, with disagreements decided by three hadith scholars, and report how often the annotators and the scholars agree.
- Report the two supporting studies of the claim with their scripts and limits.

Out of scope

- Linking narrator mentions to an authority list (entity resolution) and an identity ledger inside the engine; domain rules that need biographical dates; question answering over the graph; multi-model experiments and a labelled set larger than 50 hadiths. These are planned for Project II.
- Turkish hadith text: we found no copyright-free Turkish source with full chains.
- Judging the authenticity of hadiths. The system records chains; it does not grade them.

Success criteria

- **S1.** 100 hadiths pass from upload to canonical graph with no manual step outside the review screen (logged end-to-end run).
- **S2.** 100% of facts in the canonical graph have a source quotation and a decision record (automated test over the graph).
- **S3.** Precision, recall and F1 are reported for four conditions (rule-based parser; LLM; LLM with checks; LLM with checks and human review) on the 50 labelled hadiths, under strict and loose matching, with hadith-level bootstrap confidence intervals. For each check the table gives how many wrong proposals it flags on its own, next to what the model's confidence flags at the same review share.
- **S4.** Agreement between the two annotators (mutual F1), agreement among the three scholars (Fleiss' kappa), the number of contested items and the labelling time per hadith are reported.
- **S5.** Every number of the two supporting studies is reproduced by a script from a clean clone of the repository, and the report states what was measured for and against the claim.

4. Similar Systems and Background

System / work	What it does	Limitation or gap your project addresses
Sanadset 650K [1]	650,986 hadith records from 926 books with narrator tags produced by regular expressions	No human check of the tags

SemanticHadith [2], [3]	Ontology and knowledge graph of the six canonical collections; narrators extracted with a custom NER model (micro F1 97%); three experts checked samples	No LLM extraction and no decision per fact; ambiguous names are left for manual review; expert validation framework listed as future work
Muther and Smith [4]	Expert-annotated isnad boundaries in 50 classical texts	Finds isnad spans, not narrators; experts agree exactly on the end point in only 53% of cases
AR-Sanad 280K [5]	27,056 real chains linked to 18,298 narrator identities	Starts from separated names, not text; 60.6% of chains contain at least one wrong narrator
Oumer [6]	Fine-tuned narrator and relation extraction on 223,478 hadiths (narrator F1 0.98)	Needs large labelled training data; no provenance or human review
Rezwan [7]	LLM-assisted corpus of 1.2M narrations scored by six experts	Scores without an agreement statistic; not comparable across studies
Text2KGBench [8]	Benchmark for ontology-driven graph extraction with LLMs	Shows that "appears in the text" does not imply "correct" (precision 0.38 on one dataset)
Tsaneva et al. [10]	Nine workflows combining LLM validators and people to validate a scientific graph, scored against three experts	Whole workflows compared; single modules are not evaluated in isolation
KARMA [11]	Multi-agent LLM enrichment of a biomedical graph with validation agents	Ablation scored by an LLM judge, not an external reference
ODKE+ [12]	Production extraction with a second LLM as grounding judge	No deterministic check of the extracted value; method of the reported effects not given

Position of this project. A recent review of the field [9] names weak benchmark comparability and sparse expert validation as open problems. A hadith knowledge graph, its ontology and accurate fine-tuned extractors already exist; we build on them for the schema. Routing uncertain facts to people by confidence or by disagreement is not new [10]–[12], and we do not claim it. We did not find a check that the extracted value occurs in the model's own quotation reported as its own error class, nor a measurement of what each deterministic check catches on the same candidates against an external reference, compared with the model's confidence at the same review share. Our search was not systematic, so this is a statement about what we found.

5. Preliminary Requirements

Functional requirements

ID	Requirement	Priority (must / should / could)
FR-1	The system shall ingest a hadith as one source unit with a stable identifier and content hash.	must
FR-2	The system shall split a hadith into isnad and text and return narrator mentions with character offsets.	must
FR-3	The system shall request candidates from an LLM for one chain at a time, constrained to the schema.	must
FR-4	The system shall reject a candidate whose type is not in the schema without discarding the other candidates of the same call, and record the unknown type.	must
FR-5	The system shall mark a candidate for review when its quotation does not occur in the source.	must
FR-6	The system shall mark a candidate for review when its extracted value does not occur in its quotation.	must
FR-7	The system shall mark a candidate for review when it names the Prophet as a narrator.	should
FR-8	The system shall show candidates with their quotation and check results and let a reviewer accept, reject or edit them.	must

FR-9	The system shall record for every decision the reviewer, the time and the duration.	must
FR-10	The system shall write accepted facts to the canonical graph so that links between the same two narrators in different hadiths stay distinct.	must
FR-11	The system shall display the chain of a hadith with the source quotation of each link.	should
FR-12	The system shall score output against a reference file and report precision, recall and F1.	must
FR-13	The system shall let a user define node types, relation types and properties, and reject identifiers the graph database cannot store.	could

Non-functional requirements

ID	Requirement	How it will be checked
NFR-1	No fact reaches the canonical graph without a decision record.	Automated test (S2)
NFR-2	Re-running a commit does not create duplicate nodes or links.	Integration test
NFR-3	The backend test suite runs without a database or network access in under one minute.	CI
NFR-4	A new user can start the system from the README alone.	Test on a clean machine in week 14
NFR-5	No API keys or passwords are stored in the repository.	Repository scan in CI

6. Method and Technology Plan

Approach. The engine is a pipeline with a gate at each stage. The model has no write access to the graph; a candidate becomes a fact only after it passes the checks and receives a decision.

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hadith text → isnad segmenter → LLM proposal (per chain) → checks → review → canonical graph
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Measurement. We add the stages one at a time and score each configuration against the hand-labelled set, so the contribution of each stage can be read from the table. The matching rule is fixed before measurement: strict (normalised strings equal) and loose (equal, substring, or token F1 of at least 0.60), with maximum-weight matching between system and reference mentions. The hadith is the unit of analysis.

Labelled data and the hadith scholars. Two group members label the same 50 hadiths independently, following a written guideline and without seeing system output; the member who writes the checks does not label. Our supervisor advised working with more than one scholar, because hadith scholars can disagree among themselves. Three scholars take part: Prof. Dr. Rifat Atay (Faculty of Theology) and two scholars we will identify by week 6. All three review the guideline before labelling starts. Every item on which the two annotators disagree goes to the three scholars, who decide it independently of each other, and the majority answer enters the reference. When all three answers differ, the item is marked contested, left out of the main score and reported separately. Ten randomly chosen hadiths on which the annotators agreed also go to the scholars, to catch errors both annotators share. The scholars do not judge system output, so the people who set the reference do not also judge the system.

Existing code. GiraffeGraph Core already implements ingestion, LLM extraction, candidate validation, a review queue and commit to Neo4j. The checks and the decision record were written in October with AI coding assistants and have automated tests, but the group has not reviewed them and no reviewer has used them. Reviewing, testing and measuring them is the main work of this term.

Layer / component	Technology or tool	Reason for the choice
Backend	Python 3.12, FastAPI	Existing code base; team experience
Operational store	PostgreSQL 16	Runs, candidates, review items and decisions are relational
Canonical graph	Neo4j 5	Path queries over chains
Review dashboard	Next.js 15	Existing review interface
LLM access	OpenAI-compatible HTTP API	Switch between hosted and open models without code changes

Development	uv, pytest, ruff, Docker Compose, GitHub Actions	Reproducible setup and CI
Data	Sanadset 650K (text); Ifta Sunnah labels for measurement only	Arabic text with full chains under an open licence; platform labels are not redistributed

Use of AI assistants. AI assistants were used for the literature search, the measurements, the code and the drafting of this document. References were retrieved from Crossref, arXiv and the ACL Anthology. The group is responsible for every statement here.

7. Work Plan

WP	Work package and its output	Responsible member(s)	Weeks
WP1	Architecture, data model, three screen sketches and technology decision table	Yahya Efe Kuruçay; all for review	5–7
WP2	Labelling guideline reviewed by the three scholars; 50 hadiths labelled by two annotators; scholar decisions; agreement figures and reference set	Deniz Büyükşahin, Alperen Ulukaya	6–8
WP3	Midterm report and presentation	All	8
WP4	Group review of the existing engine; segmenter, per-chain extraction, three checks and the Prophet rule tested	Alperen Ulukaya (segmenter), Yahya Efe Kuruçay (checks)	9–10
WP5	Review screen used by a group member on the labelled set; decision record and chain view tested	Deniz Büyükşahin (review screen), Alperen Ulukaya (chain view), Yahya Efe Kuruçay (decision record)	10–11
WP6	Integration on 100 hadiths, scoring script, test against S1–S5, final report, README and demo	All; Yahya Efe Kuruçay (scoring)	12–15

Supervisor meetings follow the course plan (weeks 6–7, 9–10 and 12–13). Every member commits their own work under their own account.

8. Expected Outputs and Risks

Expected outputs

- A working prototype in one private repository with README, tests and CI; the engine is published under an open-source licence at the end of the project.
- A labelling guideline and 50 labelled hadiths with annotator and scholar agreement figures.
- A results table for the four conditions in S3.
- The two supporting studies with their scripts, result files and stated limits.
- Final report, presentation and a plan for Project II.

Risks

Risk	Likelihood	Impact	Mitigation plan
Labelling takes longer than planned	Medium	Medium	Measure time per hadith on the first ten; label in a spreadsheet instead of building a tool
Annotators disagree often	Medium	High	Add boundary rules to the guideline and repeat the round
A scholar cannot take part or is late	Medium	Medium	Work with two scholars; when they disagree the item is contested

Scholars disagree on many items	Medium	Medium	Report the contested share as a result; clarify the guideline with the scholars
Development starts late	Medium	High	Simplify the interface; keep the checks and the scoring
Licence of the platform labels cannot be confirmed	Medium	Low	Use them for measurement only; rely on Sanadset and our own labels

9. References

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[12] S. Khorshidi et al., "ODKE+: Ontology-Guided Open-Domain Knowledge Extraction with LLMs," arXiv:2509.04696, 2025.

Supervisor Approval

Supervisor name	
Assessment	Approved ☐ Approved with revisions ☐ Revise and resubmit ☐
Comments	
Date / signature	